When Can Training Data Be Recovered?

A rank condition for non-identifiability from the first-layer weight signal

A detailed, self-contained proof: if the gate matrix has rank below the number of training samples, the inputs cannot be recovered from the first-layer weight signal — no matter the algorithm.

Yoad Oxman · thesis note · 2026-08-20

In plain terms — read this first

The reconstruction attack tries to read the training images back out of a trained network's weights. This note pins down a hard limit on when that is possible even in principle.

Every training sample leaves a fingerprint in the first layer's weights. But the fingerprint is filtered through how strongly each neuron reacts to that sample — a number we call the GATE. Collect all the gates into a matrix M: one row per neuron, one column per sample. The claim is:

> The samples can be separated (recovered) only if M has rank at least N, the number of samples. If rank(M) < N, several samples share essentially the same gating pattern, their fingerprints add together into a blend, and no algorithm can un-mix them.

We (1) derive exactly what the attacker observes, (2) show it is a matrix product 'gates × data', (3) prove the rank condition step by step, (4) count exactly how many indistinguishable alternative datasets exist, and (5) exhibit two completely different datasets that produce the identical weights — which you can verify by hand.

Notation

d	input dimension (e.g. number of pixels).
N	number of training samples in the batch — the thing we count against.
m	number of hidden neurons (the width of the layer).
$x_i \in \mathbb{R}^d$	the i-th training input; the columns of the data matrix X. This is what we recover.
$w_k \in \mathbb{R}^d$	weight vector of neuron k; the rows of the weight matrix W.
$v_k \in \mathbb{R}$	output weight of neuron k (second layer).
σ, σ'	the activation function and its derivative.
$\langle w_k, x_i \rangle$	the pre-activation of neuron k on sample i (a scalar).
M_{ki}	the GATE MATRIX entry: $M_{ki} = \sigma'(\langle w_k, x_i \rangle)$.
$c_i \in \mathbb{R}$	a per-sample scalar coefficient coming from the loss (defined in §1).
Ω	the observed first-layer signal — an m×d matrix (a gradient or a weight change).

1. What the attacker sees, derived from scratch

We use the standard two-layer network of the reconstruction literature. With m hidden units and activation σ,

$$f(x;\theta) = \sum_{k=1}^{m} v_k\,\sigma(\langle w_k,\, x\rangle), \qquad w_k \in \mathbb{R}^d,\ v_k \in \mathbb{R}.$$

Given the training set $\{(x_i, y_i)\}$ and a per-sample loss ℓ, the empirical loss is $L(\theta) = \Sigma_i\, \ell(f(x_i; \theta), y_i)$. The attacker observes the FIRST-LAYER weight signal — either the gradient of the loss with respect to the first-layer weights (which, at a stationary or KKT point of training, equals a known function of the final weights), or the first-layer part of a fine-tuning weight change ΔW (one gradient step on those weights). Both have the same structure, which we now derive.

Differentiate L with respect to a single neuron's weights w_k. Writing $c_i := \ell'(f(x_i), y_i)$ for the loss slope at sample i, the chain rule gives (only the k-th term of the sum defining f depends on w_k)

$$\frac{\partial L}{\partial w_k} = \sum_{i=1}^{N} c_i \frac{\partial f(x_i)}{\partial w_k}, \qquad \frac{\partial f(x_i)}{\partial w_k} = v_k\,\sigma'(\langle w_k, x_i\rangle)\,x_i.$$

Substituting, the (k, l) entry of the observed matrix $\Omega := \partial L/\partial W$ — row k (neuron), column l (input coordinate) — is

$$\Omega_{k,l} = \sum_{i=1}^{N} c_i\,v_k\,\sigma'(\langle w_k,\, x_i\rangle)\,x_{i,l}, \quad \text{equivalently} \quad \Omega_{k,:} = \sum_{i=1}^{N} c_i\,v_k\,\sigma'(\langle w_k,\, x_i\rangle)\,x_i^{\top} \quad (\text{row } k).$$

Two facts about c_i that matter: it is a single scalar per sample, and it is the SAME for every neuron k (it does not carry a k index). For the max-margin / KKT form of the attack the identical structure appears with $c_i = \lambda_i\, y_i$ (a Lagrange multiplier times the label). Everything below only uses that c_i is a nonzero scalar shared across neurons.

> **Intuition.**
> Each neuron reports a weighted sum of the training images. The weight it puts on image x_i is its gate $\sigma'(\langle w_k, x_i\rangle)$ — how sensitive that neuron is to that image. The image only ever enters the observation through these gated sums. So the question 'can we recover the images?' becomes 'can we undo these gated sums?' — a linear-algebra question.

2. The observation is 'gates × data'

Collect the pieces into matrices. Let

$$X = [x_1,\, \dots,\, x_N] \in \mathbb{R}^{d\times N} \ \text{(columns are the samples)}, \qquad M_{ki} = \sigma'(\langle w_k, x_i\rangle) \in \mathbb{R}^{m\times N},$$

$$D_v = \operatorname{diag}(v_1,\, \dots,\, v_m), \quad D_c = \operatorname{diag}(c_1,\, \dots,\, c_N), \quad G := D_v\,M\,D_c \in \mathbb{R}^{m\times N}.$$

The two diagonal matrices merely rescale the rows and columns by the (nonzero) output weights and loss coefficients; the informative content is the gate matrix M inside G.

> **Lemma 1 (the observation factorizes).**
> The whole observation is a single matrix product of the gates and the data:
> $$\Omega = G\,X^{\top}.$$

Proof. Compare entry (k, l) of both sides. On the right,

$$(GX^\top)_{k,l} = \sum_{i=1}^{N} G_{ki}\,(X^\top)_{i,l} = \sum_{i=1}^{N} v_k M_{ki}\, c_i\, x_{i,l} = v_k \sum_{i=1}^{N} c_i\, \sigma'(\langle w_k, x_i\rangle)\, x_{i,l},$$

which is exactly the entry $\Omega_{k,l}$ from §1. Since this holds for every k and l, $\Omega = G X^\top$. ∎

Dimension check.

Every term has a clear shape, and they compose consistently (m = neurons, N = samples, d = input dimension):

- each sample $x_i \in \mathbb{R}^d$ is a d-vector; the data matrix X is d×N; so $X^\top$ is N×d.
- each $w_k \in \mathbb{R}^d$, and the first-layer weight matrix W is m×d.
- the gate $\sigma'(\langle w_k, x_i\rangle)$ is a scalar, so the gate matrix M is m×N.
- the two diagonal scalings are m×m and N×N, so G stays m×N.
- $\Omega = G X^\top$ multiplies an m×N by an N×d — the inner N's cancel — giving an m×d matrix, the same shape as W (as it must be, since Ω is the gradient with respect to W).

$$\Omega\ (m \times d)\ =\ G\ (m \times N)\ \cdot\ X^\top\ (N \times d)$$

Lemma 2 (rescaling does not change rank).

Assume (H3): every $v_k \neq 0$ and every $c_i \neq 0$. Then the two diagonal scalings are invertible, and multiplying by an invertible matrix never changes rank, so

$$\operatorname{rank}(G) = \operatorname{rank}(D_v M D_c) = \operatorname{rank}(M).$$

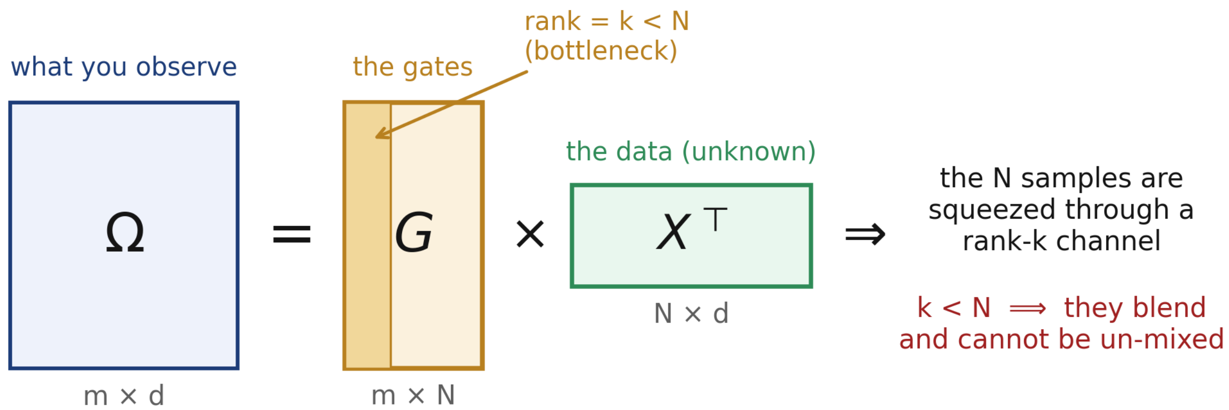

Figure 1. The observation Ω is the product of the gate matrix G (m×N, but only rank k) and the data $X^\top$ (N×d). The N samples must pass through a channel of rank k. When k < N the channel is too narrow to carry them separately — they arrive blended.

3. Three linear-algebra facts we will use

The proof needs only these standard facts; we state them plainly so the argument is self-contained.

(i) Solution sets of linear equations. If a linear map Φ satisfies Φ(X) = Ω and also Φ(X′) = Ω, then Φ(X′ − X) = 0. So every solution X′ equals the true X plus something the map sends to zero. In symbols, the full solution set is X + ker Φ, where ker Φ = {H : Φ(H) = 0} is the KERNEL (null space).

(ii) Rank–nullity. For a linear map given by a matrix A with n columns, $\dim(\ker A) = n - \operatorname{rank}(A)$. The rank counts the independent directions the map keeps; the nullity $n - \operatorname{rank}(A)$ counts the directions it destroys.

(iii) Positive dimension means infinitely many. A subspace of dimension ≥ 1 contains a whole line's worth of points — uncountably many. So if the solution set $X + \ker \Phi$ has dimension ≥ 1, there are infinitely many datasets consistent with the same observation.

> **Intuition.**
> Whatever the map 'destroys' (its kernel) is information you can never recover from the output. If the data can be changed along any kernel direction without changing Ω, then Ω simply does not determine the data. The size of the kernel is the size of your ignorance.

4. What 'recoverable' means, precisely

Fix the gate values M and the scalars v, c — i.e. treat the gates as GIVEN (assumption H2; §8 explains why this is the right and strongest way to phrase the limit). Under H2 the map from data to observation is linear:

$$\Phi : \mathbb{R}^{d \times N} \to \mathbb{R}^{m \times d}, \qquad \Phi(X) = G\,X^{\top}.$$

We say the data X is IDENTIFIABLE from Ω if the only datasets X′ with Φ(X′) = Ω are X itself and its column permutations. Column permutation is a benign symmetry — it only relabels which image we call 'the first' — so it does not count as a failure. NON-identifiability means some genuinely different dataset (not a relabeling) produces the identical observation; then no procedure, however powerful, can tell which one was the true training set.

5. The theorem and its proof

> **Theorem (rank obstruction).**
> Assume (H1) $\Omega = G X^{\top}$, (H2) the gates G are fixed, (H3) $v_k, c_i \neq 0$. If rank(M) = k < N, then the set of datasets consistent with the observation is an affine subspace
>
> $$\Phi^{-1}(\Omega) = X + K, \qquad K = \{H \in \mathbb{R}^{d \times N} : \text{ every row of } H \text{ lies in } \ker G\},$$
>
> $$\dim K \;=\; d\,(N - k) \;\geq\; d \;\geq\; 1.$$

In words: there is a whole d(N−k)-dimensional family of different datasets that all produce the same Ω. Hence X is not identifiable.

Proof, in five steps.

Step 1 (the solution set is X + kernel). Φ is linear (Lemma 1: $\Phi(X) = G X^{\top}$). By fact (i), the datasets consistent with Ω are exactly $\Phi^{-1}(\Omega) = X + \ker \Phi$, since the true X is one solution.

Step 2 (what is in the kernel?). We find every H with $\Phi(H) = G H^{\top} = 0$. Fix a column index l and look at column l of $G H^{\top}$:

$$(GH^\top)_{k,l} = \sum_{i=1}^{N} G_{ki} H_{l,i} = \left(G \cdot (\text{row } l \text{ of } H)^\top\right)_k.$$

This is zero for all neuron indices k exactly when G times (row l of H)$^\top$ is the zero vector — that is, when row l of H lies in ker G. This must hold for each of the d rows of H, and the rows are unconstrained otherwise, so

$$\ker \Phi = \{H : \text{ each of the } d \text{ rows of } H \in \ker G\}.$$

Step 3 (count the dimensions). Each of the d rows ranges independently over the space ker G, so the dimension of the kernel is d copies of dim(ker G). By rank–nullity (fact ii) applied to G, which has N columns,

$$\dim \ker \Phi = d \cdot \dim(\ker G) = d\,(N - \operatorname{rank} G) = d\,(N - k),$$

using rank G = rank M = k from Lemma 2.

Step 4 (the family is nontrivial). Because k < N we have N − k ≥ 1, so dim ker Φ = d(N−k) ≥ d ≥ 1. By fact (iii) this is a positive-dimensional space: it contains infinitely many distinct H, hence $\Phi^{-1}(\Omega)$ = X + ker Φ contains infinitely many distinct datasets, all producing the identical Ω.

Step 5 (they are genuinely different, not relabelings). The benign symmetries are the N! column permutations (plus any finite sign/scale symmetries) — a FINITE set. A finite set cannot fill a positive-dimensional continuum, so all but a measure-zero portion of the consistent datasets are unrelated to X by any symmetry. Therefore no function of Ω can single out the true X. ∎

6. A concrete example you can check by hand

Take the smallest interesting case: m = 2 neurons, N = 2 samples, input dimension d = 2. Suppose the (fixed) gate-and-coefficient matrix is

$$G = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow \text{ column } 2 = 2 \times \text{column } 1, \quad \operatorname{rank}(G) = 1 < 2 = N.$$

Let the true dataset have samples $x_1 = (1, 0)^\top$ and $x_2 = (0, 1)^\top$, i.e. X = I. The observation is

$$\Omega = G X^\top = G = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Now consider a completely different dataset $x_1' = (3, 2)^\top$ and $x_2' = (-1, 0)^\top$. Multiplying out (you can do it in your head):

$$G X'^{\top} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \Omega$$

> Two datasets with nothing in common — $\{(1,0), (0,1)\}$ versus $\{(3,2), (-1,0)\}$ — produce the IDENTICAL observation Ω. An attacker seeing Ω cannot possibly know which one was the training set. This is the theorem made concrete: with rank(G) = 1 < N = 2, the consistent datasets form a d(N−k) = 2-parameter family (here $x_1' = (1+2s, 2t)$, $x_2' = (-s, 1-t)$ for any s, t), all giving the same weights.

7. Why low rank blurs the samples (the mechanism)

When two samples have proportional gate columns (the rank-1 case), every neuron reacts to them in lockstep — whatever neuron k does to x_1, it does a fixed multiple of to x_2. The observation then depends on the two images only through one fixed linear combination of them, say $s = c_1x_1 + \gamma c_2x_2$. The individual images are free to slide along the line that keeps s fixed, and every point on that line fits the data equally well. The reconstruction can at best return the blend s — never the two images apart. This is exactly the 'superposition' failure seen empirically for N ≥ 2 once the per-sample gradients become collinear, and it is why raising the gate rank (through the activation, the anchor, or the LoRA rank) is what buys separability.

8. Corollary, scope, and honest caveats

> **Corollary (a necessary width).**
> Recovering X from the first-layer signal requires rank(M) ≥ N. Since rank(M) ≤ min(m, N), it in particular requires the hidden width m ≥ N: a layer narrower than the batch cannot expose its samples through this signal, whatever the activation, optimizer, or algorithm.

(a) Why 'fixed gates' is the right framing. In reality the gates depend on the data ($M_{ki} = \sigma'(\langle w_k, x_i\rangle)$), so the true map $X \mapsto \Omega(X) = G(X)\, X^{\top}$ is bilinear, and the alternative datasets X′ generally have different gates. The theorem therefore describes the STRONGEST possible attacker — one additionally handed the exact gate values of the true data. A real attacker knows less, so cannot do better on this count. What is rigorously proved: the first-order (gate-factorization / NTK-linearized) information in Ω is INSUFFICIENT when rank(M) < N, and that first-order regime is exactly where the attack operates. The extra self-consistency the true gates must satisfy can in special cases shrink the family, so this is a necessary condition, not a claim that no nonlinear trick could ever help.

(b) Necessary vs. sufficient. rank(M) ≥ N is necessary. In the fixed-gate model it is also generically sufficient: if rank(G) = N then $X = (G^{+}\Omega)^{\top}$ is the unique solution (G^{+} = Moore–Penrose pseudoinverse). Turning this into a clean sufficiency theorem for the full bilinear problem — exactly when ΔW determines $\{x_i\}$ uniquely — is the open identifiability question the thesis targets.

(c) One layer. We used the first-layer factorization, where the data appears explicitly

($\partial L/\partial W_1$ carries x_i^T). Deeper layers may add constraints; the bound is a lower bound on difficulty, tight in the linearized regime.

9. Why this matters for the thesis

rank(M) is a hard ceiling on how many samples can be separated from the first-layer signal, and the three levers of the study act on it directly. The ACTIVATION determines σ' and hence the entries of M. The ANCHOR — linearizing partway between the initial weights θ_0 and the fine-tuned weights (the α-sweep below) — shifts the pre-activations $\langle w_k, x_i\rangle$ and so reshapes M. A LoRA adapter replaces Ω by a rank-r projection of it, tightening the ceiling further to

$$\text{leakage} \;\le\; \min(\text{rank}(M),\; r,\; N), \qquad \theta_{\text{anchor}} = (1-\alpha)\,\theta_0 + \alpha\,\theta_T.$$

The theorem above is the first, information-theoretic, of these ceilings — the reason 'how many samples, and through what activation and rank' is the right axis for the whole study.